

Pharmaniaga Gliclazide Tablet

DESCRIPTION

White, round, flat, bevelled-edge tablet with Pharmaniaga icon on one side and intersected breakline on the other.

COMPOSITION

Each tablet contains Gliclazide 80 mg.

PHARMACODYNAMICS

Gliclazide is a hypoglycaemic sulphonylurea of 2nd generation which lowers blood glucose by directly stimulating the acute release of insulin from functioning beta cells of pancreatic islet tissue by unknown process that involves a sulphonylurea receptor on the beta cell. Sulphonylureas inhibit the ATP-potassium channels on the beta cell membrane and potassium efflux, which results in depolarisation and calcium influx, calcium-calmodulin binding, kinase activation, and release of insulin-containing granules by exocytosis, an effect similar to that of glucose. Insulin is a hormone that lowers blood glucose and controls the storage and metabolism of carbohydrates, proteins and fats. Therefore, sulphonylureas are effective only in patients whose pancreas are capable of producing insulin.

In addition to the pancreatic action, it has been demonstrated that gliclazide administration improves the metabolic utilisation of glucose at a peripheral level. This extrapancreatic action is not due to a modification in the number of insulin receptors, but may be due to the potentiation of the post-receptor pathways.

Thus, gliclazide restores glycaemic control throughout 24 hours. It normalises fasting and postprandial blood sugar.

In man, apart from having hypoglycaemia effect, gliclazide has been shown to reduce platelet hyperadhesiveness and hyperaggregation and to increase fibrinolytic activity. (These factors are thought to be implicated in the pathogenesis of the long-term complications of diabetes mellitus). Controlled studies in patients with diabetic retinopathy suggest that these effects of gliclazide are beneficial in slowing down the progression of retinopathy.

PHARMACOKINETICS

Gliclazide is well absorbed from GI tract and peak plasma concentrations occur 2-4 hours after administration.

Gliclazide is extensively (85%) bound to plasma proteins. Metabolism is extensive and all the metabolites are devoid of hypoglycaemic activity. 60 - 70% of the dose is excreted in the urine, <5% is excreted unchanged in the urine and 10 - 20% in the faeces as metabolites. The elimination half-life of gliclazide is 10-12 hours.

INDICATIONS

Non-insulin - dependent diabetes mellitus.

CONTRAINDICATIONS

Gliclazide should not be used in insulin-dependent diabetes, pregnancy, diabetics complicated by ketosis and acidosis, diabetics undergoing surgery, diabetic coma, diabetic pre-coma, hyperosmolar nonketotic coma, after severe burns or trauma or during infections and patients known to have hypersensitivity to other sulphonylureas and related drugs. It is also contraindicated in severe renal and hepatic insufficiency.

ADVERSE REACTIONS

Gastro-intestinal disturbances such as nausea, vomiting, heartburn, anorexia, diarrhoea and a metallic taste may occur with sulphonylureas and are usually mild and dose-dependent. Increased appetite and weight gain may occur. Skin rashes and pruritus may occur and photosensitivity has been reported. Rashes are usually hypersensitivity reactions and may progress to more serious disorders. Hypoglycaemia occurs with all hypoglycaemic agents.

Other severe effects may be manifestations of a hypersensitivity reaction. They include cholestatic jaundice, leucopenia, thrombocytopenia, aplastic anaemia, agranulocytosis, haemolytic anaemia, erythema multiforme or the Steven-Johnson syndrome, exfoliative dermatitis and erythema nodosum.

The sulphonylureas may infrequently induce a syndrome of inappropriate secretion of antidiuretic hormone (SIADH) characterised by water retention, hyponatraemia and central nervous system effects.

PRECAUTIONS

Paediatric

Safety and efficacy of oral antidiabetics agents have not been established in paediatrics.

Risk benefit should be considered when the following medical problems exist:

Severe diarrhoea, severe gastroparesis, intestinal obstruction, prolonged vomiting, hyperglycaemia, hyperthyroidism, hypoglycaemia, hypothyroidism and renal function impairment. Use of sulphonylureas increases the risk of possibly prolong hypoglycaemia with renal function impairment.

Risk benefit should also be considered in hepatic disease, sulphonylureas that are extensively metabolised in the liver should not be used when there is hepatic impairment. Hypoglycaemia that develops may be more severe when these are being used.

Note:

Hypoglycaemia - Severe hypoglycaemia may occur in any patient treated with any sulphonylurea. The propensity to cause hypoglycaemia does not appear to be solely related to the half life of the drug. So caution must always be taken when taking any sulphonylureas especially in renal or hepatic function impairment.

Adjustment of dosage of hypoglycaemic agents may be required in patients suffering from intercurrent infections, traumas, shock or after anaesthesia. When major surgery is to be performed, insulin therapy should be substituted for oral hypoglycaemics.

Pregnancy and Lactation

Pregnancy

Studies in humans have not been done. Gliclazide is not recommended for use during pregnancy.

Breast Feeding

Gliclazide should not be used during breast feeding.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Care is necessary during excessive exercise especially while driving and during the use of machine because hypoglycaemia may be provoked. If hypoglycaemia may occur, any moving action should be stopped until recovery is complete.

DRUG INTERACTIONS

Alcohol:

A disulfiram - like reaction, which is characterised primarily by flushing of the face, neck and arms may occur with any of the sulphonylureas when alcohol is ingested concurrently. The risk of hypoglycaemia may be increased or prolonged when moderate or large amounts of alcohol have been consumed concurrently with sulphonylurea antidiabetic agents.

Anticoagulants (coumarin - or indandione derivative) will increase their anticoagulant and hypoglycaemic effects.

Potential of the hypoglycaemic action of gliclazide may occur with the concomitant administration of sulphonamides, salicylates, phenylbutazone, β -adrenoreceptor-blocking agents, MAOIs, ketoconazole and miconazole. Diminution of hypoglycaemic action of gliclazide may occur with the concomitant administration of thiazide diuretics, corticosteroids and oestrogens.

DOSAGE AND ADMINISTRATION

DOCTORS ARE ADVISED TO TITRATE THE REQUIRED DOSAGE ACCORDING TO INDIVIDUAL PATIENT'S RESPONSE DURING INITIAL ADMINISTRATION.

The dose should be adjusted according to the individual patient's response, commencing with 40 - 80 mg daily and increasing until adequate control is achieved. A single dose should not exceed 160 mg and when higher doses are required, a twice-daily split dosage is advised and should be divided according to the main meals of the day. In the majority of cases, 2 tablets/ day with meals, (1 tablet with breakfast, 1 tablet with dinner).

For the elderly diabetic, the starting dose should be ½ tablet in the morning and ½ tablet in the evening.

ROUTE OF ADMINISTRATION

Oral.

OVERDOSAGE

The symptoms to be expected of overdose would be hypoglycaemia. The treatment is gastric lavage and correction of the hypoglycaemia by appropriate means with continued monitoring of the patient's blood sugar until the effect of the drug has ceased.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

SHELF LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

In blisters of 60 and 100 tablets.

Date of Revision: 12th September 2024

PRODUCT REGISTRATION HOLDER/ MANUFACTURER:

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